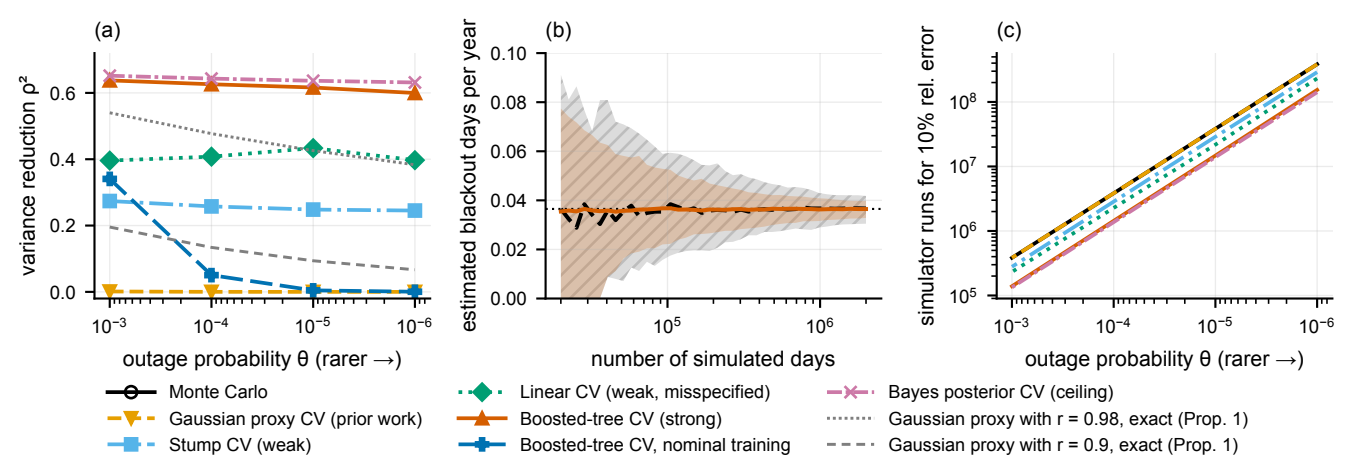
(a)
variance reduction ρ²
0.0
0.2
0.4
0.6
10⁻³
10⁻⁴
10⁻⁵
10⁻⁶
outage probability θ (rarer →)
(b)
estimated blackout days per year
0.00
0.02
0.04
0.06
0.08
0.10
10⁵
10⁶
number of simulated days
(c)
simulator runs for 10% rel. error
10⁵
10⁶
10⁷
10⁸
10⁻³
10⁻⁴
10⁻⁵
10⁻⁶
outage probability θ (rarer →)
Monte Carlo
Gaussian proxy CV (prior work)
Stump CV (weak)
Linear CV (weak, misspecified)
Boosted-tree CV (strong)
Boosted-tree CV, nominal training
Bayes posterior CV (ceiling)
Gaussian proxy with r = 0.98, exact (Prop. 1)
Gaussian proxy with r = 0.9, exact (Prop. 1)